SOBEY SCHOOL OF BUSINESS
MASTER OF BUSINESS ANALYTICS
ACCOUNTING & FINANCE ANALYTICS (MBAN5570)
ASSIGNMENT - 1

Instructor: Mohammad M. Rahaman
Lecture Location: SB415
Meeting Time: W: 2.00PM-5.00PM

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DUE DATE: January 21, 2026 (Brightspace Dropbox)

Please read the following instructions carefully: This is your first assignment. This assignment would be done exclusively using Excel and Python. For some of the questions, you will need specific data set. I will give you all the relevant information here. Only one copy of your write-up should be submitted. Report your results clearly, and omit your computational details if they are not asked for. Explain briefly when necessary. Please do not bury your answers in large tables of numbers. Type all your answers (whether numerical or not) in a few pages. When you submit, make sure that you also give me your .pdf files only. Do not hand in un-edited output. Submit in the designated Dropbox in Brightspace. You can download all relevant data files for this assignment from the Brightspace Course Page. The assignment will be graded out of 10 and all questions are equally weighted.

I. Descriptive Analytics of Financial Data

In this exercise, you are asked to do a simple descriptive analytics of a financial variable. First, log on to your WRDS account. Go to Compustat - Capital IQ. Click on North America - Daily. Go to Security Monthly. In the Query Form choose the following: Step 1: 2010-01 to 2024-12; Step 2: TIC and put MSFT in the company code; Step 3: Select Company Name (CONM) and PRCCM – Price – Close – Monthly; Step 4: Excel spreadsheet (*.xlsx), none, DDMMYY10). Then hit Submit Query. Once you follow this process, you will get the Microsoft monthly closing stock price from Jan 1, 2010 until December 31, 2024. You are free to use either Excel or Python for the next step.

1. Winsorzie the raw price data at the 1% level (with and without replacement) and perform a simple descriptive analyses of the MSFT price data. Next, winsorzie the raw price data at the 5% level (with and without replacement) and perform another simple descriptive analyses of the MSFT price data. What level and type of winsorization is most appropriate for your MSFT stock price data.

2. After you decide on the level and type of winsorization, nexrt I want to you to work with the transformed data instead of the raw data that you collected. Calculate the return of MSFT stock price using the following formula: $\frac{P_t - P_{t-1}}{P_{t-1}}$, where P_t is the MSFT stock price in period t. Perform a simple descriptive analytics of the return data and comment on whether the return data needs further winsorization.

3. Based on whatever software you use, suggest the appropriate distributions for the MSFT monthly stock price and the monthly return data. Please submit a graph of the PDF of the distributions and highlight the key parameters of the distribution.

4. Now take the logarithm transformation of MSFT stock price and suggest an appropriate distribution for the logarithm of MSFT stock price. Please submit a graph of the PDF of the distribution of the logarithm of MSFT stock price.

5. If you wish to perform further diagnostic, predictive, and prescriptive analytic on the MSFT stock price, which data would be the most suitable one: raw MSFT stock price data, winsorized MSFT stock price data, logarithm of the MSFT stock price data, or the return of the MSFT stock price data. Please explain your reasoning as clearly and succinctly as possible.

II. Financial Data Visualization & Mining

Collecting Stock Price Data: In this exercise you will analyze the stock price data of Appple inc. I want you to collect the daily stock price data for the company from January 2010 to December 2024. Here are the steps to collect stock price data:

1. Apple Inc ticker symbol is; AAPL.

2. Use an APIs from Yahoo Finance to collect the Apple Inc stock price data from January 2010 to December 2024.

3. Save the stock price data in a Pandas DataFrame.

Analyzing Stock Price Data: After collecting the stock price data, the next step for you is to analyze the price information as follows:

1. Plot a line graph to show the adjusted close price for any month of your choice. You can choose any month from January 2010 to December 2024 or you can chose the entire series.

2. Plot the daily returns of the stock price data for that month.
3. Plot the cumulative daily return of the stock price data for that month.
4. Calculate the volatility of the stock price data for that month. Note that, in finance and accounting we calculate volatility using the statistical measure of standard deviation.
5. Comment on your findings.

Moving Average Crossover Trading Strategy: In this task, you will use the moving average crossover strategy to find out what time frames from January 2010 to December 2024 were good times to buy the stocks. This is also an example of financial data mining, where you deploy an arbitrary trading rule in a set of historical financial data to identify what rules can potentially generate profitable trading opportunities in the future. Here are the steps you need to follow:

1. Create two window periods, one short-term for 50 days and one long-term for 200 days.
2. Calculate the moving average for the short-term and long-term window periods.
3. Create a Signal column that takes the value of 1 when the short-term moving average is above the long-term moving average, and -1 when the short-term moving average is below the long-term moving average.
4. Identify the buy and sell signals based on the Signal column.
5. Plot the results.

It is important to note that this exercise may require some additional research and reference to the tutorials provided in the course. Additionally, the final exercise should be submitted in a report format that includes detailed explanations of each step, as well as any code and visualizations used. You can use Google Colab notebook to complete this exercise, as it provides an efficient and accessible way to write and run code in a browser-based environment. You should include text blocks where appropriate to provide context and explanation for their code, as well as to describe their findings and observations throughout the exercise. Please submit a PDF report of your analyses in the dropbox. You do not need to submit any Python codes.

III. AI-Assisted Financial Analytics and Critical Evaluation

In this exercise, you will use artificial intelligence (AI) tools to replicate and extend the analyses you conducted in Question 1 (MSFT via WRDS) and Question 2 (AAPL via Yahoo Finance and Python). The purpose of this question is not to replace your prior work, but to critically evaluate AI as an analytical assistant relative to your own traditional analyses. You are free to use any large language model (LLM) or AI-enabled analytics tool (e.g., ChatGPT, Claude, Gemini, Copilot, or similar). However, you remain fully responsible for the correctness, interpretation, and judgment of all outputs.

Part A: AI-Generated Analysis (Replication and Extension): Using the same datasets from Question 1 (MSFT monthly prices and returns) and Question 2 (AAPL daily prices and trading strategy), ask an AI tool to perform analyses comparable to those you have already completed. At a minimum, instruct the AI to do the following:

1. Perform descriptive and diagnostic analysis of MSFT stock prices and returns, including discussion of:
   - Outliers and winsorization,
   - Distributional properties,
   - Appropriate transformations (log prices, returns).
2. Analyze AAPL stock price data and generate:
   - Data visualizations of prices and returns,
   - Volatility discussion,
   - A simple moving average (50–200 day) crossover trading strategy.
3. Ask the AI to interpret the results, not just compute them (e.g., what the patterns imply about risk, trends, or trading behavior).

You should clearly document: What prompts you used (summarized, not verbatim), What tasks you delegated to AI, What outputs the AI generated.

Part B: Benchmarking AI Against Your Own Analysis: Using your answers from Question 1 and Question 2 as the benchmark, critically compare the AI-generated analyses with your own work. Your discussion must explicitly address the following:

1. What did the AI do well? Identify areas where the AI analysis:
   - Was faster, clearer, or more systematic,
   - Identified patterns or insights you did not initially emphasize,
   - Produced useful visualizations or summaries.

2. What did the AI do poorly or incorrectly? Discuss whether the AI:
   - Misinterpreted financial concepts,
   - Made incorrect statistical assumptions,
   - Oversimplified risk, distributions, or trading logic,
   - Failed to recognize data limitations.
3. Which AI insights do you accept—and why? Clearly identify which AI-generated outputs you incorporated into your understanding of MSFT or AAPL, and justify your acceptance using finance, accounting, or statistical reasoning.
4. Which AI insights do you reject—and why? Explicitly state what you discarded, and explain whether this was due to:
   - Conceptual errors,
   - Weak economic intuition,
   - Inconsistency with your empirical results.

Part C: Impact on Your Financial Assessment: Reflect on how (if at all) your assessment of MSFT and AAPL changes after incorporating AI insights.

1. Did AI alter your view of:
   - The distributional properties of returns?
   - The usefulness of transformations (winsorization, logs, returns)?
   - The credibility or limitations of simple technical trading rules?
2. If your assessment changed, explain how and why.
3. If it did not change, explain why your original analysis remains superior.

Part D: Reflection on AI as an Analytics Tool: Conclude with a short reflection:

1. Under what circumstances does AI enhance financial analytics?
2. Where does AI require human judgment and domain knowledge to avoid errors?
3. How would you use AI going forward as a complementary tool—not a substitute—for financial analysis?